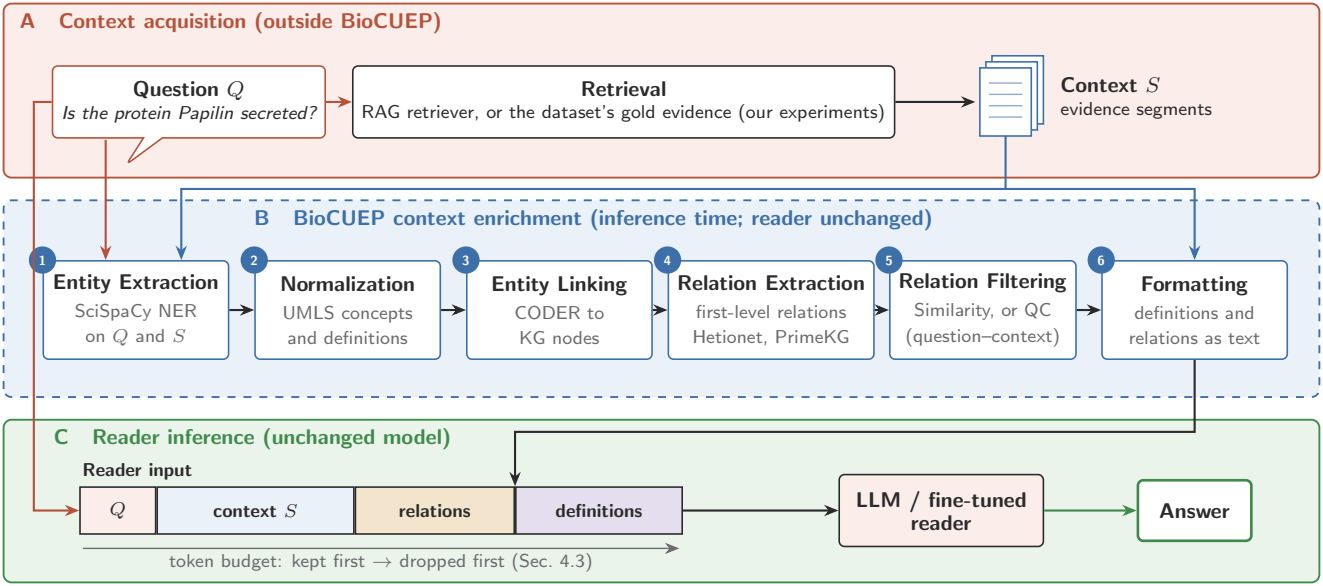

A Context acquisition (outside BioCUEP)
Question Q
Is the protein Papilin secreted?
Retrieval
RAG retriever, or the dataset's gold evidence (our experiments)
Context S
evidence segments
B BioCUEP context enrichment (inference time; reader unchanged)
1
Entity Extraction
SciSpaCy NER on Q and S
2
Normalization
UMLS concepts and definitions
3
Entity Linking
CODER to KG nodes
4
Relation Extraction
first-level relations Hetionet, PrimeKG
5
Relation Filtering
Similarity, or QC (question–context)
6
Formatting
definitions and relations as text
C Reader inference (unchanged model)
Reader input
Q
context S
relations
definitions
token budget: kept first → dropped first (Sec. 4.3)
LLM / fine-tuned reader
Answer